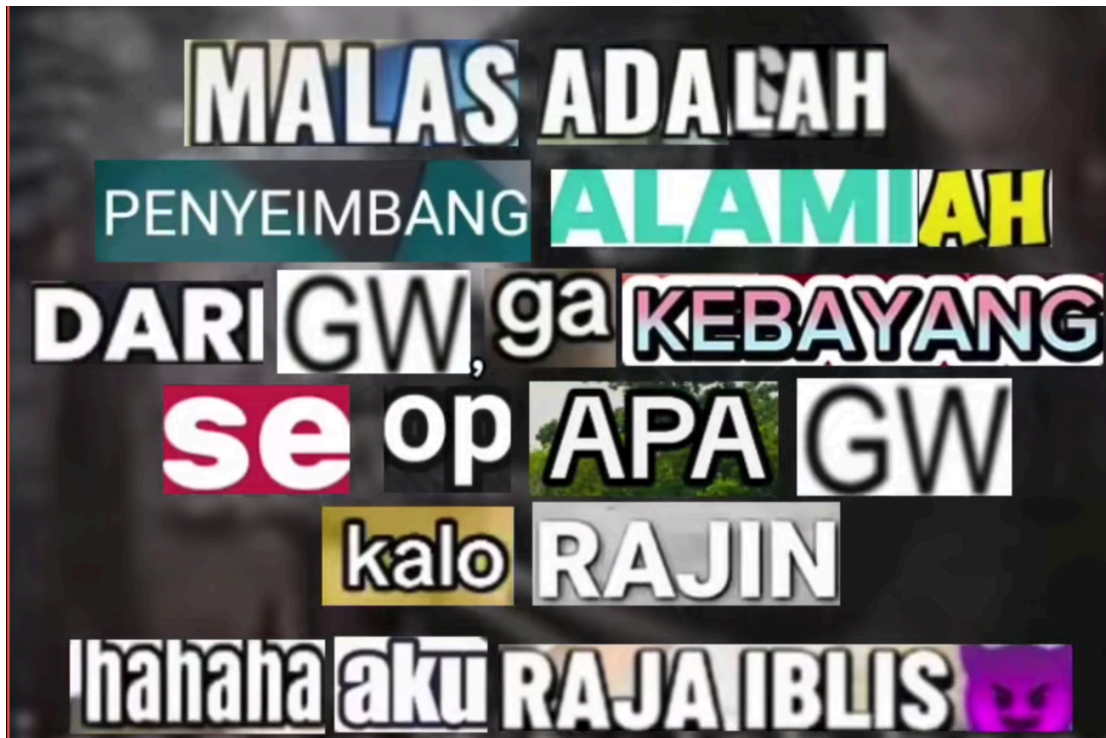


WRITE UP TUGAS WEEK 1



BY:

Ngurah Gde Wisnu Kertha Murthi

DAFTAR ISI

1. GENERAL SKILL

- 1.1 Strings It
- 1.2 Log Hunt
- 1.3 Running Binary
- 1.4 Binhexa
- 1.5 Multi Code
- 1.6 Nice Netcat
- 1.7 Super SSH
- 1.8 Piece by Piece

2. Tools, script dan command yang digunakan

3. Flag yang berhasil tercapture

1.1 Strings It

```

-----
ultron@cachyos-x8664
-----
OS: CachyOS x86_64
Host: 1701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 4 hours, 3 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
Desktop Environment: GNOME 50.4
Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4x8) @ 3.30 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 8.14 GiB / 15.36 GiB (53%)
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Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4x8) @ 4.40 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 8.15 GiB / 15.36 GiB (53%)
-----
cd Downloads/

~/Downloads
cd tugasi_week1 1/

~/Downloads/tugas week 1
strings strings

```

Pertama tama, aku download filenya, lalu aku buka filenya

```

$3412
$6332
$2632
$3250
$3215
$2200
$4178
$4139
$4674
$8744
$8374
.symtab
.strtab
.shstrtab
.interp
.note.gnu.property
.note.gnu.build-id
.note.ABI-tag
.gnu.hash
.dynsym
.dynstr
.gnu.version
.gnu.version_r
.rela.dyn
.rela.plt
.init
.plt.got
.plt.sec
.text
.fini
.rodata
.eh_frame_hdr
.eh_frame
.init_array
.fini_array
.dynamic
.data
.bss
.comment

~/Downloads/tugas week 1 - fish
~/Downloads/tugas week 1
$ strings strings | grep "pico"
picoCTF{StRiNg5_1t_47948C73}
~/Downloads/tugas week 1
$
```

```

Sep 13 10:58 PM
~/Downloads/tugas week 1- fish

ultron@cachyos-x8664
-----
OS: CachyOS x86_64
Host: 1701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 1 hour, 16 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
Desktop Environment: GNOME 50.4
Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4+8) @ 3.30 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 4.80 GiB / 15.36 GiB (31%)

ultron@cachyos-x8664
-----
OS: CachyOS x86_64
Host: 1701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 1 hour, 16 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
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GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 4.80 GiB / 15.36 GiB (31%)

> cd Downloads/

~/Downloads
> cd tugas1_week1/

~/Downloads/tugas week 1
> strings server_log

```

```
Sep 13 10:58 PM
~/Downloads/tugas week 1 - fish

[1990-08-09 13:33:30] DEBUG System check complete
[1990-08-09 13:33:38] WARN Disk space low
[1990-08-09 13:33:47] ERROR Connection lost

~/Downloads/tugas week 1
> strings server.log | grep "pico"
[1990-08-09 10:00:10] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:04:27] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:04:29] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:04:37] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:19:23] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:19:29] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:19:32] INFO FLAGPART: picoCTF{us3_

~/Downloads/tugas week 1
> strings server.log | grep "INFO FLAGPART"
[1990-08-09 10:00:10] INFO FLAGPART: picoCTF{us3_
[1990-08-09 10:02:55] INFO FLAGPART: y0urlinux_
[1990-08-09 10:05:54] INFO FLAGPART: skills_
[1990-08-09 10:05:55] INFO FLAGPART: skills_
[1990-08-09 10:10:54] INFO FLAGPART: cedfa5fb)
[1990-08-09 10:10:58] INFO FLAGPART: cedfa5fb)
[1990-08-09 10:11:06] INFO FLAGPART: cedfa5fb)
[1990-08-09 11:04:27] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:04:29] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:04:37] INFO FLAGPART: picoCTF{us3_
[1990-08-09 11:09:16] INFO FLAGPART: y0urlinux_
[1990-08-09 11:09:19] INFO FLAGPART: y0urlinux_
[1990-08-09 11:12:40] INFO FLAGPART: skills_
[1990-08-09 11:12:45] INFO FLAGPART: skills_
[1990-08-09 11:16:58] INFO FLAGPART: cedfa5fb)
[1990-08-09 11:16:59] INFO FLAGPART: cedfa5fb)
[1990-08-09 11:17:00] INFO FLAGPART: cedfa5fb)
[1990-08-09 12:19:23] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:19:29] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:19:32] INFO FLAGPART: picoCTF{us3_
[1990-08-09 12:23:43] INFO FLAGPART: y0urlinux_
[1990-08-09 12:23:45] INFO FLAGPART: y0urlinux_
[1990-08-09 12:23:53] INFO FLAGPART: y0urlinux_
[1990-08-09 12:25:32] INFO FLAGPART: skills_
[1990-08-09 12:28:45] INFO FLAGPART: cedfa5fb)
[1990-08-09 12:28:49] INFO FLAGPART: cedfa5fb)
[1990-08-09 12:28:52] INFO FLAGPART: cedfa5fb)

~/Downloads/tugas week 1
>
```

Lalu saya grep pico dengan command `strings server.log | grep "pico"` Flagnya ketemu tapi kepotong, jadi saya run command `strings server.log | grep "INFO FLAGPART"` dan ketemu potongan potongan flagnya, lalu ketemu flagnya yang dalam kondisi terpecah, lalu disusun dan berhasil

2.3 Running Library


```
nc titan.picoctf.net 53719

Welcome to the Binary Challenge!
Your task is to perform the unique operations in the given order and find the final result in hexadecimal that yields the flag.

Binary Number 1: 10100110
Binary Number 2: 00001111

Question 1/6:
Operation 1: '|'
Perform the operation on Binary Number 1&2.
Enter the binary result: 302
Incorrect input. Provide the right input
Enter the binary result: 106
Incorrect input. Provide the right input
Enter the binary result: 15
Incorrect input. Provide the right input
Enter the binary result: 10101111
Correct!

Question 2/6:
Operation 2: '*'
Perform the operation on Binary Number 1&2.
Enter the binary result: 00001111
Incorrect. Try again
Enter the binary result: 100110111010
Correct!

Question 3/6:
Operation 3: '&'
Perform the operation on Binary Number 1&2.
Enter the binary result: 00000110
Correct!

Question 4/6:
Operation 4: '+'
Perform the operation on Binary Number 1&2.
Enter the binary result: 179
Incorrect input. Provide the right input
Enter the binary result: 181

Operation 4: '+'
Perform the operation on Binary Number 1&2.
Enter the binary result: 179
Incorrect input. Provide the right input
Enter the binary result: 181
Correct!

Question 5/6:
Operation 5: '>>'
Perform a right shift of Binary Number 2 by 1 bits .
Enter the binary result: 1111000
Incorrect. Try again
Enter the binary result: 10001110
Incorrect. Try again
Enter the binary result: 10001111
Incorrect. Try again
Enter the binary result: 10000111
Incorrect. Try again
Enter the binary result: 00000111
Correct!

Question 6/6:
Operation 6: '<<'
Perform a left shift of Binary Number 1 by 1 bits.
Enter the binary result: 110100111
Incorrect. Try again
Enter the binary result: 01001101
Incorrect. Try again
Enter the binary result: 01001100
Incorrect. Try again
Enter the binary result: 01001101
Incorrect. Try again
Enter the binary result: 01001100
Incorrect. Try again
Enter the binary result: 101001100
Correct!

Enter the results of the last operation in hexadecimal: 14C
Correct answer!
The flag is: picoCTF{01tw^3se_0p3eR@t10n_su33essFul_d5f8047e}
```

Dapatkan netcat dari web CyLab, lalu aku mengakses nc (netcat) di terminal dan diberikan soal

1. Operation |, artinya or, jika ada 1 dan 0/1, maka outputnya adalah 1, jika 0 dan 0, maka outputnya adalah 0

- Habis semua operation itu, bilangan binary terakhir di translate ke hexadecimal, lalu flag berhasil diperoleh

```

Sep 14 12:02 AM
~/Downloads/tugas week 1 - fish

-----
() OS: CachyOS x86_64
Host: l701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 2 hours, 20 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
Desktop Environment: GNOME 50.4
Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4x8) @ 4.40 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 7.79 GiB / 15.36 GiB (51%)

ultron@cachyos-x8664
-----
() OS: CachyOS x86_64
Host: l701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 2 hours, 20 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
Desktop Environment: GNOME 50.4
Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4x8) @ 3.30 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [Integrated]
Memory: 7.79 GiB / 15.36 GiB (51%)

cd Downloads/

~/Downloads
cd tugas1_week1_1/

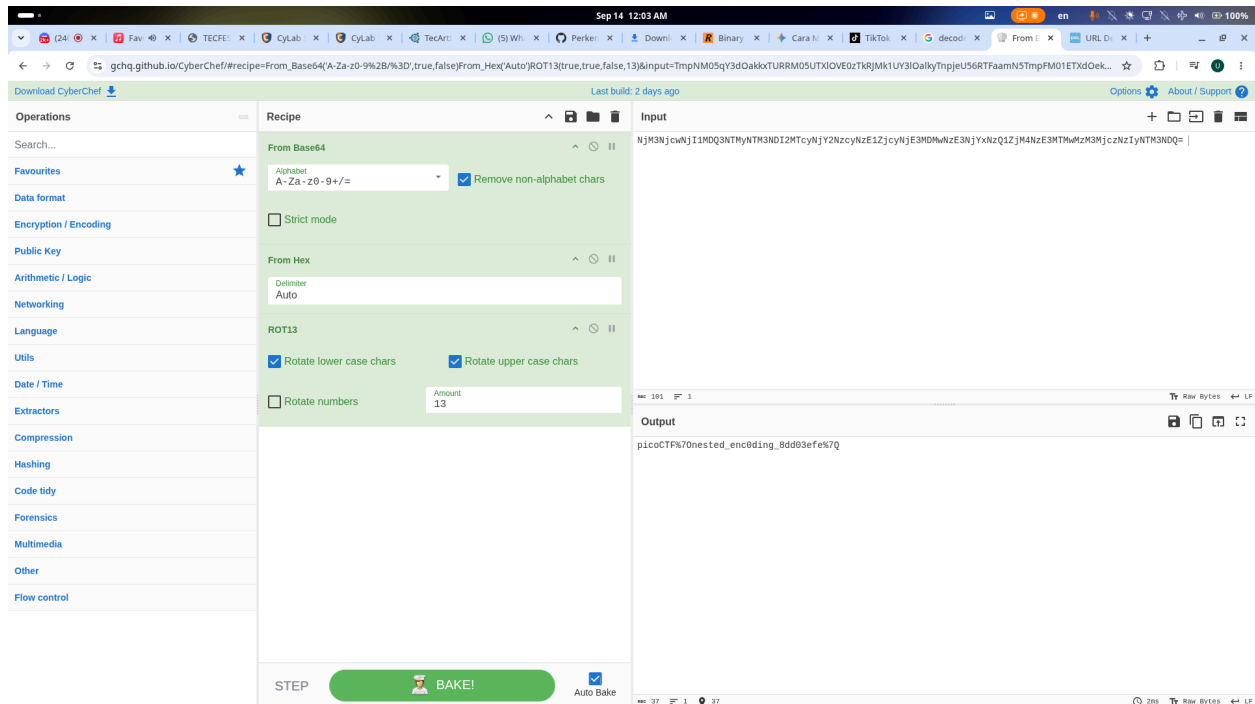
~/Downloads/tugas week 1
file message.txt
message.txt: ASCII text

~/Downloads/tugas week 1
strings message.txt
MjM3NjcwNjE1MDQ3NTMyNTM3NDI2MTcyNjY2NzcyZmE1ZjcyNjE3MDQwMzE3NjYxNzQ1ZjM4NzE3MTMwMzQ3MjczNzI1NTM3NDQ=

~/Downloads/tugas week 1

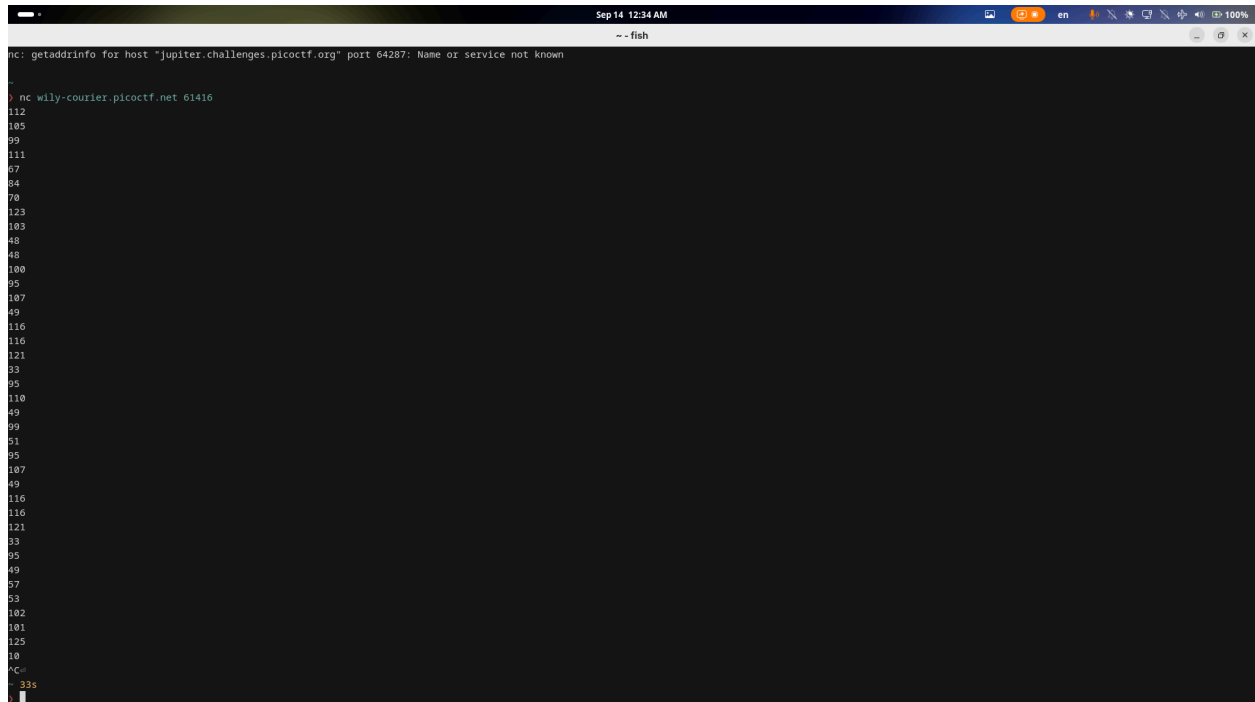
```

Saya coba download filenya, lalu saya strings filenya dan keluar hasil, ada kode panjang



lalu saya coba translate dengan website CyberChef di chrome, lalu saya masukan operation From base64 karena kalimat dari file message.txt memiliki = diakhirnya, lalu saya masukan operation From hex karena dari hasil From base64, keluar output angka yang sangat banyak, lalu saya masukan operation ROT13 berdasarkan hint yang diberikan dari website CyLab, dan flagnya berhasil tercapture, namun ada catatan, karena error hasil output akhirnya, saya ubah sendiri %70 menjadi { dan %7Q menjadi }.

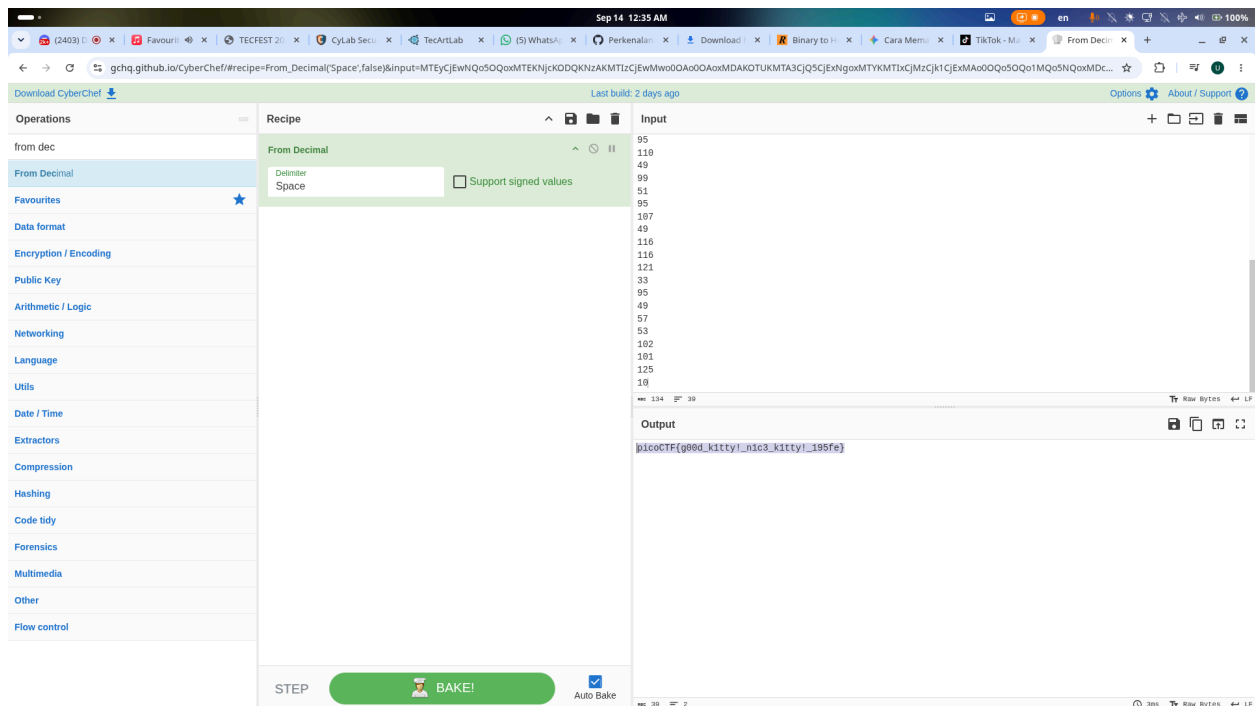
2.6 Nice Netcat



```
nc: getaddrinfo for host "jupiter.challenges.picoctf.org" port 64287: Name or service not known

nc wily-courier.picoctf.net 61416
112
105
99
111
67
84
78
123
103
48
48
100
95
107
49
116
116
121
33
95
110
49
99
51
95
107
49
116
116
121
33
95
49
57
53
102
101
125
18
10
335
```

aku mengakses netcat yang diberikan dari CyLab, aku mendapatkan barisan angka



lalu saya translate dengan CyberChef, dengan From Decimal, dan flagnya ketemu

2.7 Super SSH

```
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
ssh ctf-player@titan.picoctf.net -p 50414

yes
f3be1b38
^C
- 6m 7s
> ssh
usage: ssh [-46AaCfGgKkMnNqsTtVvXxyYz] [-B bind_interface] [-b bind_address]
          [-c cipher_spec] [-D [bind_address]:port] [-E log_file]
          [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
          [-j destination] [-L address] [-l login_name] [-m mac_spec]
          [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
          [-S ctl_path] [-w host:port] [-w local_tun[:remote_tun]]
          destination [command [argument ...]]
          ssh [-Q query_option]

> ssh -p 50414
usage: ssh [-46AaCfGgKkMnNqsTtVvXxyYz] [-B bind_interface] [-b bind_address]
          [-c cipher_spec] [-D [bind_address]:port] [-E log_file]
          [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
          [-j destination] [-L address] [-l login_name] [-m mac_spec]
          [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
          [-S ctl_path] [-w host:port] [-w local_tun[:remote_tun]]
          destination [command [argument ...]]
          ssh [-Q query_option]

> ssh -p 50414 ctf-player@titan.picoctf.net
The authenticity of host '[titan.picoctf.net]:50414 ([3.139.174.234]:50414)' can't be established.
ED25519 Key fingerprint is: SHA256:459EBTSSR2m32I+c0M5TyztHpQryvSkudRP9P1K7XQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[titan.picoctf.net]:50414' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
ctf-player@titan.picoctf.net's password:
Permission denied, please try again.
ctf-player@titan.picoctf.net's password:
Welcome ctf-player, here's your flag: picoCTF(s3cur3_c0nn3ct10n_3e293eeea)
Connection to titan.picoctf.net closed.

- 29s
>
```

Aku mendapatkan instance di CyLab, lalu membuka terminal dan mengetik `ssh -p ctf-player@titan.picoctf.net`, lalu ada konfirmasi untuk lanjut terkoneksi, pilih yes, lalu ada disuruh masukan password yang sudah diberikan di instance, Flag berhasil tercapture

2.8 Piece by Piece

```
ultron@cachyos-x86_64
-----
OS: CachyOS x86_64
Host: 1701
Kernel: Linux 7.2.3-1-cachyos
Uptime: 3 hours, 27 mins
Packages: 12 (flatpak), 1297 (pacman)
Shell: fish 4.9.1
Desktop Environment: GNOME 50.4
Window Manager: Mutter (Wayland)
CPU: 12th Gen Intel(R) Core(TM) i3-1220P (4+8) @ 3.30 GHz
GPU: Intel UHD Graphics @ 1.10 GHz [integrated]
Memory: 7.58 GiB / 15.36 GiB (49%)

> ssh -p 59111 ctf-player@dolphin-cove.picocftf.net
The authenticity of host '[dolphin-cove.picocftf.net]:59111 ([3.13.34.175]:59111)' can't be established.
ED25519 key fingerprint is: SHA256:rdvWHF7KlwuIElvysXTh/FjeqFI0dSEK5gaeIaW9t8
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[dolphin-cove.picocftf.net]:59111' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
ctf-player@dolphin-cove.picocftf.net's password:
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 7.0.0-1012-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ctf-player@pico-challs
```

Saya login ke ssh yang didapat dari CyLab, disini diketahui ada beberapa file dari ssh, lalu saya gabungkan pecahan file tersebut dengan command `cat *`, lalu saya gunakan command `cat part_* > flag.zip`, lalu saya unzip `flag.zip` dan masukan password yang sudah diberikan dari hint saat command `cat *` dijalankan, Ketika saya unzip, saya mendapatkan file `flag.txt` dan saya gunakan command `cat flag.txt` untuk mendapatkan flagnya, dan saya berhasil capture flagnya

1. Software & Tools

- Linux Terminal / CLI — Environment utama untuk mengeksekusi command-line interface.
- OpenSSH Client (`ssh`) — Digunakan untuk koneksi remote shell ke server CTF.
- Netcat (`nc`) — Digunakan untuk menghubungkan terminal ke port/server interaktif.
- CyberChef — Web tool dekoding data (digunakan untuk operasi *From Base64*, *From Hex*, *From Decimal*, dan *ROT13*).
- Web Browser — Digunakan untuk mengakses CyLab/PicoCTF platform dan CyberChef.

2. Command Line & Shell Utilities

- `strings` — Membaca dan mengekstrak string/teks yang dapat dibaca dari file biner atau log (`server.log`, biner, dll.).
- `grep` — Menyaring dan mencari baris teks spesifik (contoh: `grep "pico"`, `grep "INFO FLAGPART"`).
- `chmod` — Mengubah izin akses file agar dapat dieksekusi (`chmod +x warm`).
- `ls` — Menampilkan daftar file dalam direktori.
- `cat` — Membaca isi file serta menggabungkan beberapa pecahan file menjadi satu (`cat part_* > flag.zip`).
- `unzip` — Mengekstrak arsip terkompresi berpassword (`unzip flag.zip`).

3. Operasi Biner & Logika Komputasi (Binhexa)

- Bitwise & Arithmetic Operations:
 - `|` (OR)
 - `&` (AND)
 - `*` (Perkalian biner/desimal)
 - `+` (Penjumlahan biner)
 - `>>` (Right Shift)
 - `<<` (Left Shift)
- Konversi / Translation: Binary, Decimal, Hexadecimal, dan ASCII.

Flag yang berhasil tercapture

1. Strings It: picoCTF{5tRIng5_1T_47948C73}
2. Log Hunt: picoCTF{us3_y0urlinux_sk1lls_cedfa5fb}
3. Running Binary: picoCTF{b1scu1ts_4nd_gr4vy_ac5832c}
4. Binhexa: picoCTF{b1tw^3se_0p3eR@tl0n_su33essFuL_d6f8047e}
5. Multi Code: picoCTF%7Onested_enc0ding_8dd03efe%7Q
6. Nice Netcat: picoCTF{g00d_k1tty!_n1c3_k1tty!_195fe}
7. Super SSH: picoCTF{s3cur3_c0nn3ct10n_3e293eea}
8. Piece by Piece: picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_78b76e61}